Heart Disease Prediction

Shrimp Tank

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Data Preprocessing Techniques

Data Cleaning & Integrity

- **Target Standardization:** The target variable **Heart Disease** was converted from text (yes and no) to binary integers (1 and 0) to make it compatible with mathematical loss functions.
- **Identifier Removal:** The **id** column was dropped as it contains unique identifiers with no predictive power; keeping it would cause the model to "memorize" specific rows rather than learn patterns.
- **Handling Missing Values:** Rows with missing target values were removed to ensure the model was only trained on verified ground-truth data.

Feature Engineering Pipelines:

We utilized a **ColumnTransformer** to apply specific logic to different data types:

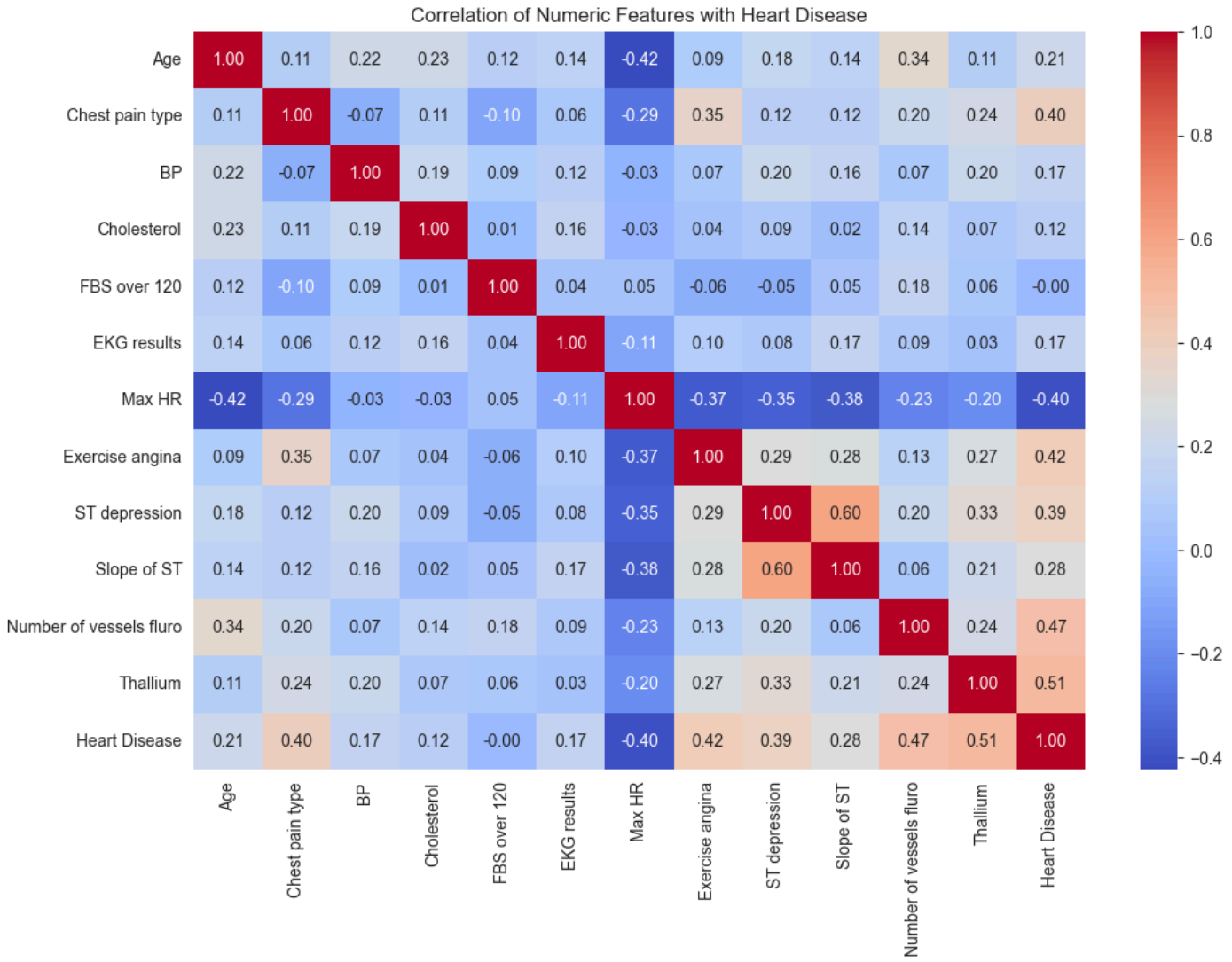
- **Numeric Features:**
    - **KNNImputer:** Instead of simple averages, we used K-Nearest Neighbors (k=5) to fill missing values based on the most similar patient profiles.
    - **StandardScaler:** Scaled data to a mean of 0 and variance of 1. This is crucial for models like SVM and Logistic Regression, which are sensitive to the magnitude of features.
- **Categorical Features:**
    - **SimpleImputer:** Filled missing categories with the "most frequent" value.
    - **OneHotEncoder:** Converted text categories (like **work_type**) into binary columns. we used **drop='if_binary'** to prevent redundancy in gender and **handle_unknown='ignore'** to ensure the pipeline doesn't crash on new data.

Exploratory Data Analysis & Visualizations

Correlation Analysis

The heatmap reveals how numeric features relate to the presence of heart disease. Key observations include:

- **Strong Positive Correlations: Thallium** (0.51), **Number of vessels fluro** (0.47), and **Exercise angina** (0.42) show the strongest positive link to heart disease.
- **Strong Negative Correlation: Max HR** (-0.40) indicates that lower maximum heart rates achieved during exercise are often associated with higher disease risk.
- **Multicollinearity Check: ST depression** and **Slope of ST** show a correlation of 0.60, suggesting they provide similar diagnostic information.



Modeling and Hyperparameters

To find the most robust predictor, we tested four distinct algorithms using **Repeated Stratified K-Fold Cross-Validation.** The scoring metric prioritized **Recall,** as missing a sick patient (False Negative) is more dangerous than a False Positive in a clinical setting.

Model	Key Hyperparameters Used
SVM	kernel='rbf', C=0.9, gamma='scale'
XGBoost	learning_rate=0.1, max_depth=3, n_estimators=100
Random Forest	n_estimators=200, max_depth=None
Logistic Regression	C=10, penalty='l2', k_best=8

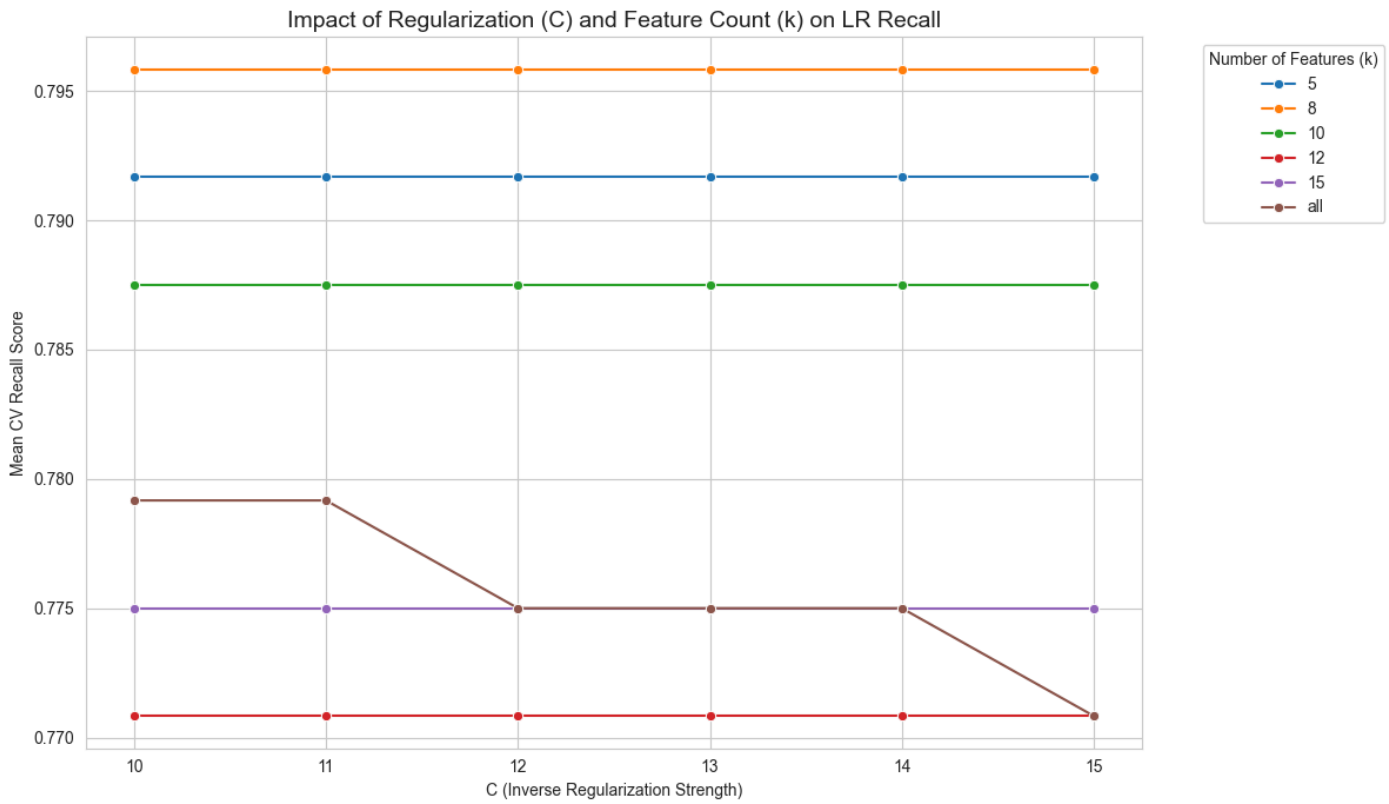
Enhancement Techniques

- **Feature Selection:** We integrated **SelectKBest** using the f-classif score within the pipeline to automatically select only the most impactful features.
- **Grid Search:** Every model underwent a **GridSearchCV** to find the optimal balance between bias and variance.
- **Class Stratification:** We used stratified splitting to ensure the proportion of healthy vs. sick patients remained consistent across training and testing sets.

Hyperparameter Sensitivity

The line plot analyzes the performance of the **Logistic Regression** model during grid search. It tracks how changing the **Inverse Regularization Strength (C)** and the **Number of Features (k)** affects the model's ability to detect sick patients (Recall).

- We observed that k=8 features provided the most stable and highest recall, while the model became less effective as more noisy features (k=15 or "all") were added.



Streamlit Application Deployment

The final **Logistic Regression** model was serialized into a **.pkl** file and deployed as an interactive web app.

- **User Interface:** Built with Streamlit, allowing clinicians to input patient metrics such as blood pressure, cholesterol, and EKG results.
- **Clinical Explanation Logic:** The app does not just give a "Yes/No" answer. It uses the model's coefficients to calculate **Diagnostic Drivers**.
- **Dynamic Feedback:** It calculates the specific probability of disease and highlights which factors (e.g., "High Thallium" or "Age") are the primary contributors to that specific patient's risk level.

Conclusion

The analysis confirms that heart disease prediction is most accurate when focusing on a subset of high-impact clinical features rather than the entire dataset. The **Logistic Regression** model performed exceptionally well, offering a balance of high recall and interpretability. By integrating feature selection and robust preprocessing into a single pipeline, we created a tool that is not only statistically sound but also practically useful for medical screening via the Streamlit interface.